

DNA Curvature and Transcription

- Clues and consequences of DNA bending in transcription

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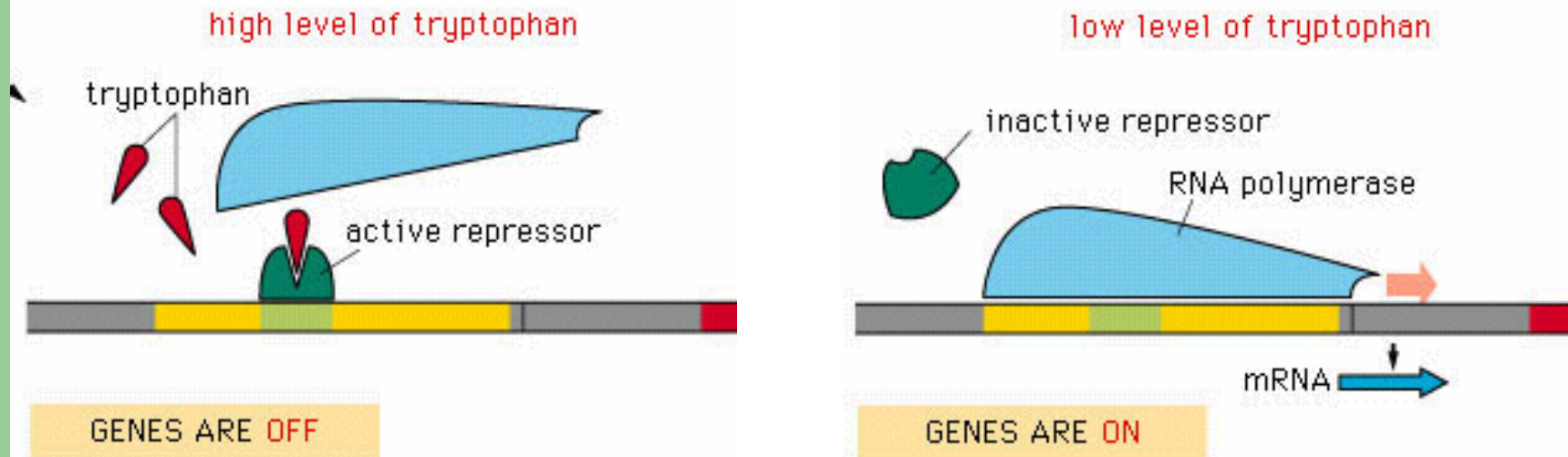
Euk vs Prok Transcription

- Several RNA Polymerases
- More general transcription factors
 - Euk. Pol can not recognize promoter regions
- Different DNA regulatory elements
 - TATA Box
 - Response Elements
- Distance of regulators from the gene
- **DNA structures are different – state of template**

Euk vs Prok

Eubacterial DNA	Eukaryotic DNA
Mostly fully exposed state and accessible as a template	Masked by organization into chromatin (and nucleosome)
Default state is transcriptionally active	Default state is to be unavailable for transcription
Regulatory mechanisms emphasize negative control through repressors to diminish the inherent activity	Regulation emphasizes activation as an essential process needed to overcome an inactive template .

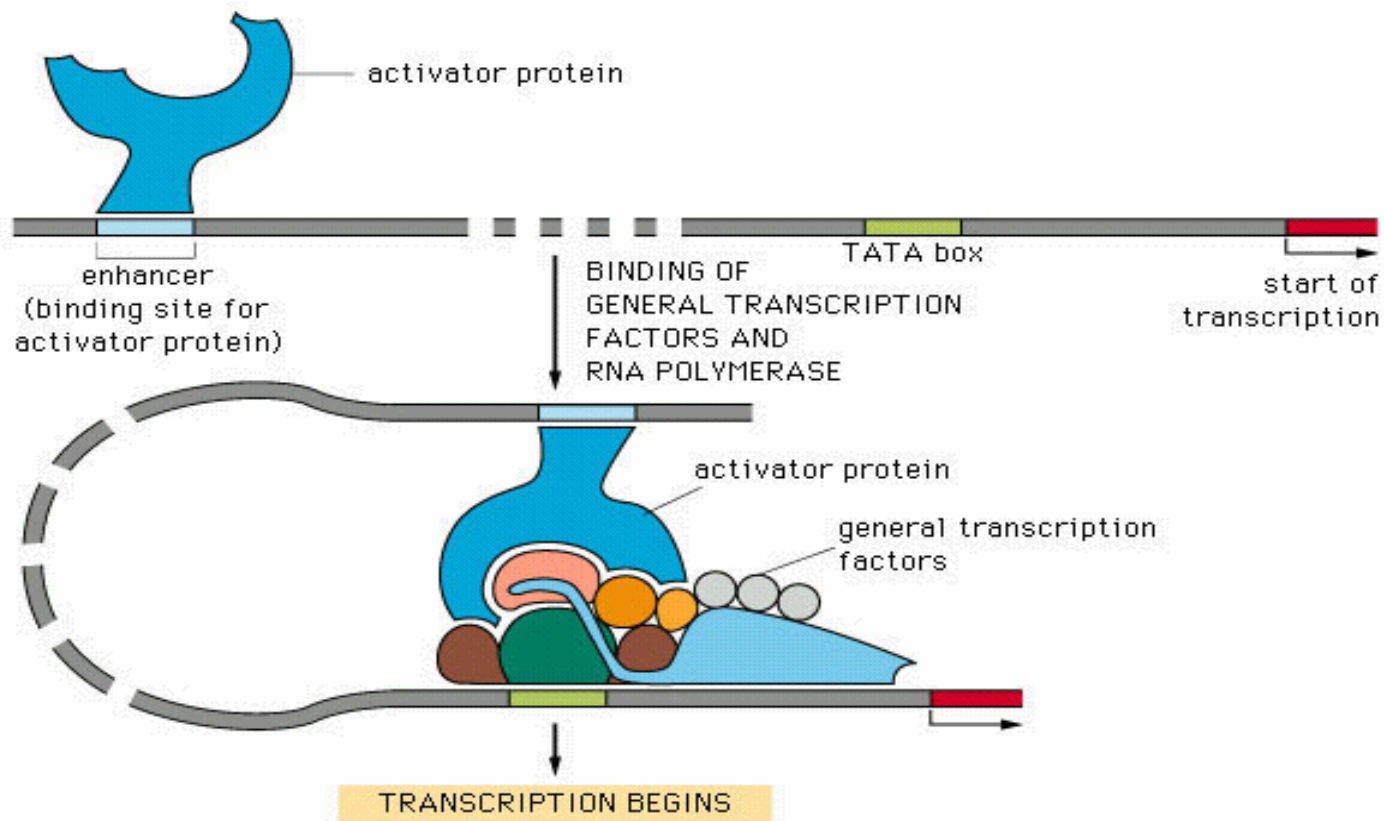
Prokaryotic Transcription Regu.



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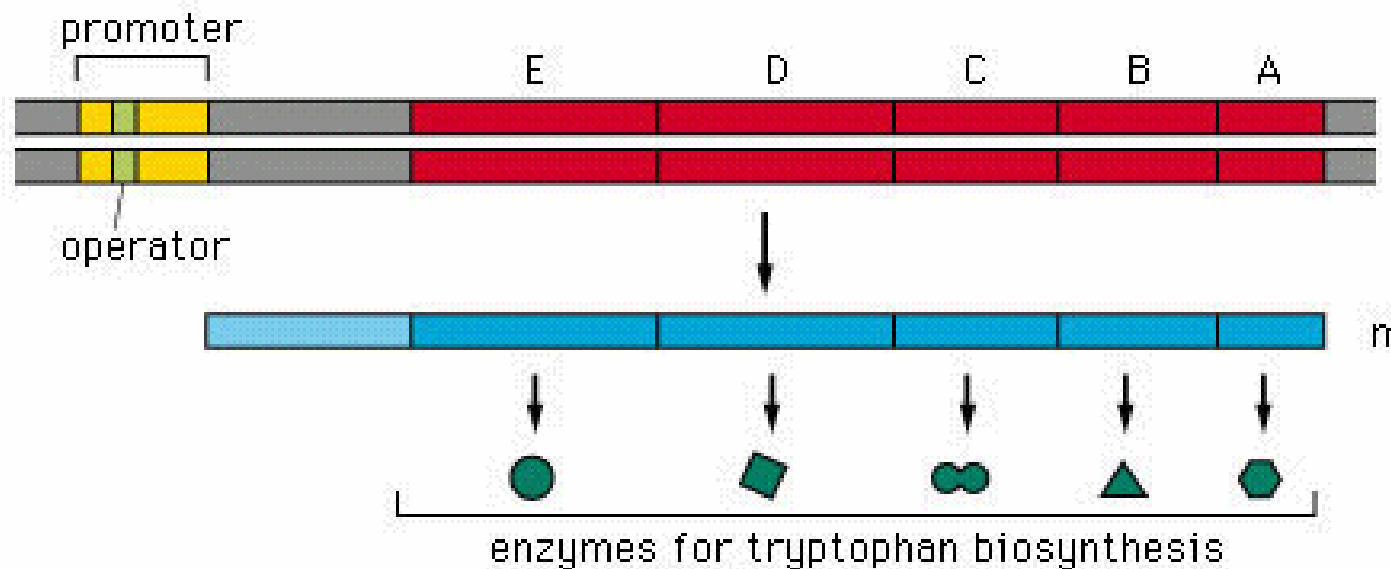
Genes expression may also be controlled with Activator proteins

Eukaryotic Transcription Regu.



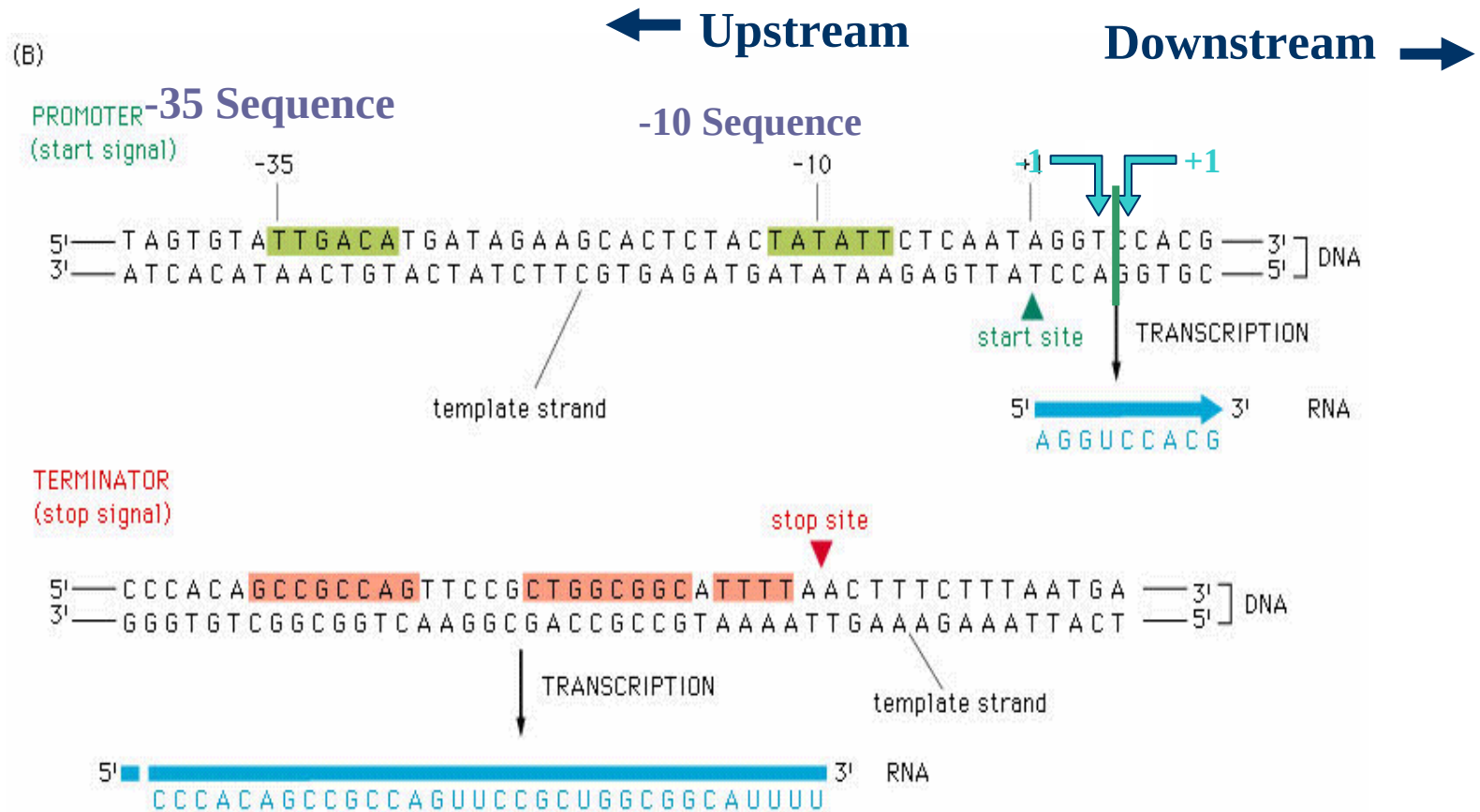
Introduction

- In prokaryotes many related genes are under the control of the same promoter



The Tryptophan operon

Introduction



Introduction

- Operators?
 - Sequences that bind a regulatory protein
- 2 types
 - repressor: bind and turn off transcription
 - activator: bind and turn on transcription
- Tryptophan operon is regulated by a repressor

Question

- Prokaryotic can have **different α -subunit** of RNA polymerase that can recognize different promoter regions
- Regulatory proteins that binds to repressor or activator
- But still much less complicated and refined
- Then how?
 - ➔ **DNA structure** can be regulatory element

Bent DNA

- **Usually found around / at**

- Origins of DNA replication
- Promoters
- Enhancers
- DNA replication loci

=> DNA curvature is of functional importance in many basic processes.

How to activate

- Bent DNA facilitate binding of RNA polymerase to promoter and/or the formation of an open promoter complex
 - **Bent DNA** : usually **right-handed**
 - **Wrapping** by Pol: **Left-handed**
- Binding of RNA Pol → change superhelical writhe from right to left → Structural strain → deformation of DNA double strand helix → unwinding → facilitating transcription

Introduction

- **Nonlinear DNA structure**
 - => **prokaryotic** cells to evolve complex signal integration device
 - => parallel the transduction cascades in higher organism to control cell growth and differentiation

Introduction

- Consequence of DNA bending in transcription
 - **Promoter geometry** becomes key regulatory element
 - Alteration of promoter architecture -> multiple metabolic control
 - Amplification and translation of small physiological signals into major transcriptional responses
 - the control of promoter specificity

The Nature of DNA Bending

- Any short homopolymeric run of **A or T** => detectable degree of curvature : **Major determinant**
- In addition to AA, AG, CG, GA, or GC can induce DNA curvature
- Poly A/T : Rigid. Bending occurs at intervening sequences or boundaries.

Bendability

- Intrinsic DNA bends $\leftrightarrow$ Bendability
 - **Curved DNA** $\rightarrow$ Rigid and deform even without external force
 - **Bendable DNA** $\rightarrow$ Many different conformational states. Establishing **equilibrium between states** depending on external forces.
- Static structures are not sufficient to explain DNA-protein recognition events.

Specificity

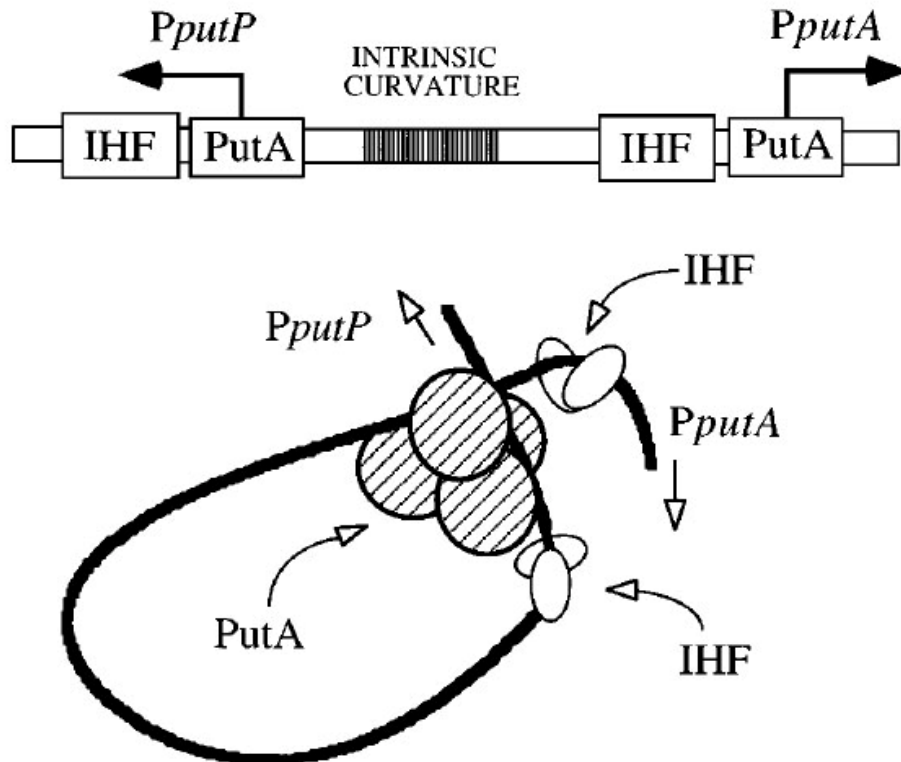
- The specificity of DNA-protein interaction
 - **Direct recognition:** Base – amino acids
 - Amino acid chain <> hydrogen atoms by the bases on the major/minor grooves
 - **Indirect readout:** Array of interaction btw protein and **DNA surface** no particular seq.
 - Amino acid chain <> sugar-phosphate

Clues of DNA Bending

- **Structural Synergy**

- Intrinsic or protein-induced bends may synergy or antagonize the interaction with proteins required to bend their targets for optimal binding.
- Ex) CAP (catabolite gene activator protein) binding site / Intrinsic bend
 - Same direction -> increase twice
 - Opposite direction -> Impairing

Clues of DNA Bending



- **E. Coli**
 - PutA : Repressor
 - IHF : Integration host factor
 - Facilitating binding of repressor to the promoter
- <- Absence of proline

DNA bending as a Catalyst of PPI

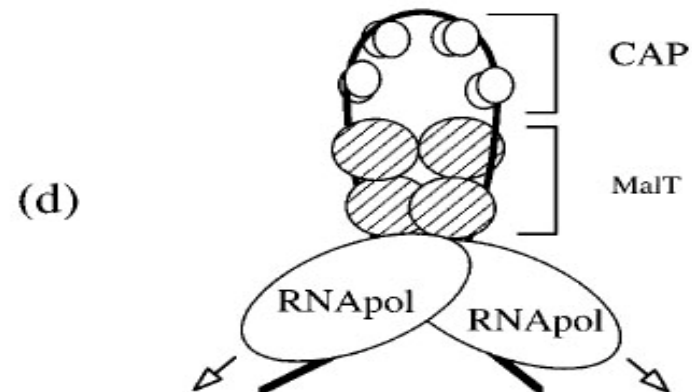
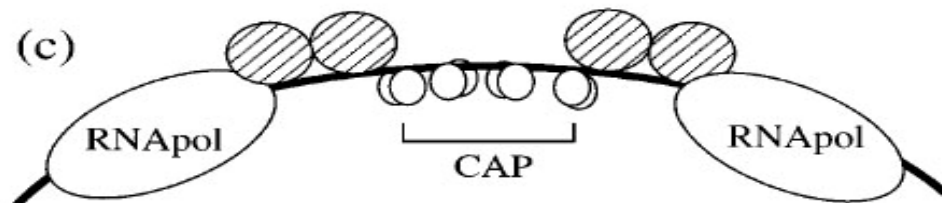
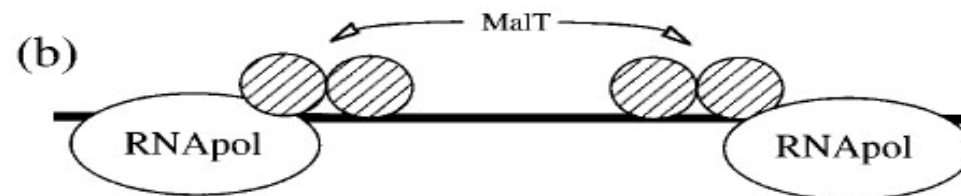
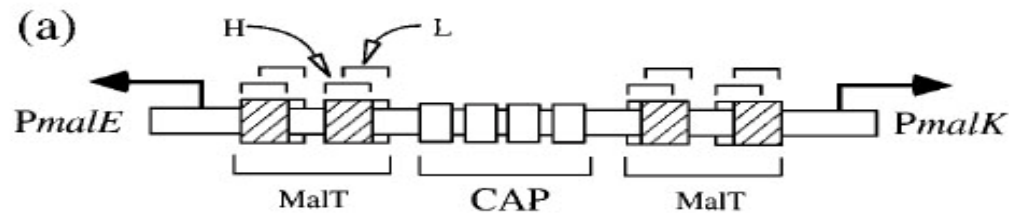
- In prokaryotic promoters, interaction btw nonadjacent DNA-bound proteins requires the formation of a loop, which is energetically disfavored if binding sites are separated by **30-140** nucleotides. => limited flexibility => Solutions?

Solutions to limited flexibility

- Altering the conformation of DNA
 - Sequence between sites may already have curved or bendable flexibility.
- Participation of extra factors
 - Either bend DNA or increase its flexibility

Consequences of DNA Bending

- Channeling Signals Through Promoter Architecture
 - Protein acting as architectural elements
=> Sustain or inhibit an active promoter configuration
 - * Protein-induced DNA bending =>
superimposition of **different regulatory levels** for the control of a single promoter



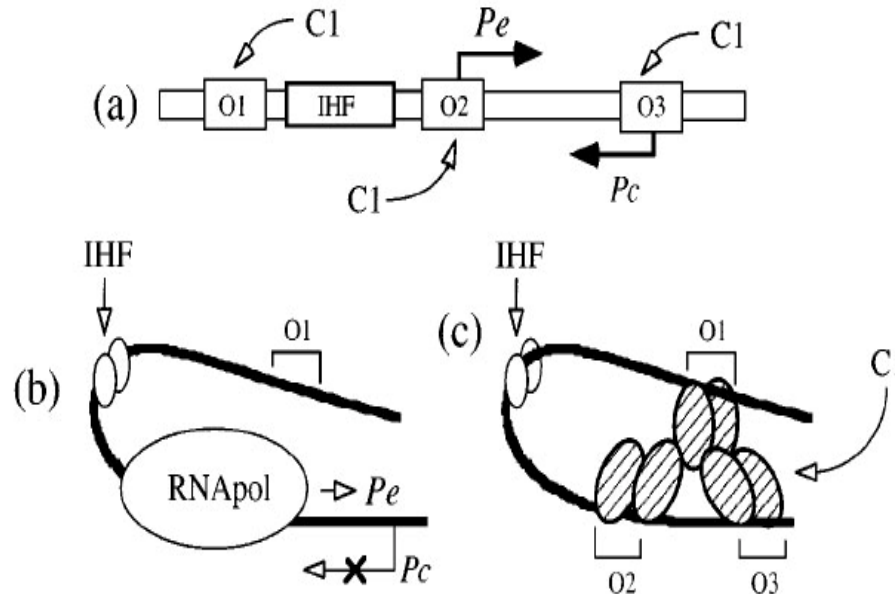
- Maltose Operon of *E. Coli*
- CAP sites can be replaced by IHF-binding sites or intrinsically curved DNA

Transcriptional Promiscuity

- Enhancer works at a distance, in different orientation, and even located downstream
 - **Same enhance may act on various promoters**
 - Degree of non-specificity is an evolutionary asset for adaptation to novel environments
 - Nonspecific activation of many promoters would be deleterious
 - **Cells need to suppress or at least control transcriptional noise**

Response-Amplification Mech.

- Minor environmental signals into major transcriptional responses.



P_e and P_c
promoters of
phage Mu

lysis (P_e) $\leftrightarrow$
lysogeny (P_c)
decision

Summary

- Static (intrinsic) and protein-induced DNA bending
→ prokaryotic cells to evolve complex signal integration devices for environmental inputs by changing **promoter architecture**.
 - Promoter geometry becomes an authentic regulatory assets
 - Multiple metabolic control levels
 - Amplifying physiological signals
 - In/decrease promoter specificity or promiscuity for cognate regulators